

Conditional inference on the asset with maximum Sharpe ratio

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Abstract

We apply the procedure of Lee *et al.* [15] to the problem of performing inference on the signal-noise ratio of the asset which displays maximum sample Sharpe ratio over a set of possibly correlated assets. We find a multivariate analogue of the commonly used approximate standard error of the Sharpe ratio to use in this conditional estimation procedure. We also consider several alternative procedures, including the simple Bonferroni correction for multiple hypothesis testing, which we fix for the case of positive common correlation among assets, the chi-bar square test against one-sided alternatives, Follman's test, and Hansen's asymptotic adjustments. [27, 6, 10]

Testing indicates the conditional inference procedure achieves nominal type I rate, and does not appear to suffer from non-normality of returns. The conditional estimation test has low power under the alternative where there is little spread in the signal-noise ratios of the assets, and high power under the alternative where a single asset has high signal-noise ratio. Unlike the alternative procedures, it appears to enjoy rejection probabilities monotonic in the signal-noise ratio of the selected asset, and actually maintains near-nominal rejection rates under the conditional null.

1 Introduction

The problem of overfitting quantitative investment strategies is certainly as old as the problem of selecting quantitative investment strategies. The choice of a course of action (*e.g.*, making an investment) based on historical observations leads to biased estimates of the value of the selected course of action when one uses the same historical observations to estimate value. That is, the estimates are “biased by selection”. This problem is not unique to quantitative finance, and goes by many names: overfitting, p-hacking, data-mining bias, *etc.* To be clear we are interested in the case where one has observed n independent contemporaneous observations of returns from k different “assets” (these can be trading strategy backtest returns, or mutual fund returns, *etc.*), selects one of those assets based on the historical performance, say by selecting the asset with maximum Sharpe ratio; then one wishes to estimate or perform inference

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on the true ‘value’ of the asset, for example its signal-noise ratio, which we define as population analogue of the Sharpe ratio.

Aronson gives a good overview of the problem from the practitioner’s point of view, noting the relevant factors are the length of history, the number of strategies tested, the correlation of their historical performance, presence of outliers (or fat-tailedness of returns), and variation in expected true effect size. [2, Chapter 6] White’s Reality Check was a pioneering development in the area, giving a generally applicable method of estimating whether a selected model was superior to a benchmark model. [33] White’s work was extended and generalized by Romano and Wolf, Hansen, *inter alia*. [26, 9, 11] From a practical point of view the Reality Check and its variants do not scale computationally to hundreds or thousands of assets, as they are based on a (block) bootstrap. However, these methods can be adapted to very general problems, can deal with correlation and autocorrelation of asset returns, and are fairly robust to assumptions.

Recent work by López de Prado and Bailey, adapting standard techniques from Multiple Hypothesis Testing (MHT), has gained attention in the field¹. [18] They find the asymptotic expected value of the maximum Sharpe ratio of uncorrelated assets with zero signal-noise ratio. While use of simple techniques from MHT (Bonferroni correction, say) can lead to reduced power, and is fragile with respect to assumptions, they are alluring in their simplicity. The Bonferroni correction is very simple to describe and implement, and does not require one to store the historical returns of the assets. It easily scales to millions of tested assets.

In this paper we exploit a result from Lee *et al.* on the problem of *conditional estimation*. [15] The Lee procedure was originally devised for analysis of the Lasso, but is applicable in general to the case of selection from a normally distributed vector conditional to a linear constraint. We simply give a multivariate normal approximation to the vector of Sharpe ratios of k assets, then appeal to the Lee *et al.* procedure.

Our procedure is midway between the simple MHT correction and the Reality Check tests, both computationally and in robustness. Our procedure requires one to estimate the correlation between returns, which would appear to require $\mathcal{O}(nk^2)$ runtimes. However, only the correlation of the selected asset against all others is required, reducing the burden to $\mathcal{O}(nk)$. Unlike the bootstrap tests, our procedure is easily adapted to the case of producing confidence intervals on the signal-noise ratio, instead of only supporting hypothesis testing.

2 Conditional Inference on the signal-noise ratio

We consider the following problem: one has observed n *i.i.d.* samples of some k -vector $\mathbf{x}$, representing the returns of k different “assets,” which could be stocks, trading strategies, *etc.* From the sample one computes the Sharpe ratio of each asset, resulting in a k -vector, $\hat{\boldsymbol{\zeta}}$. One will then choose the asset with maximum Sharpe ratio. One then seeks to perform hypothesis tests or compute confidence intervals on the signal-noise ratio of that asset. Here we define the Sharpe ratio as the sample mean divided by sample standard deviation, and the signal-noise

¹Though application of MHT corrections to the problem is not new: White’s starting assumption was apparently that such simple corrections were inadequate. [33]

ratio as the population analogue. Throughout we use hats to denote sample quantities estimating population parameters.

To simplify the exposition, we will suppose that, conditional on observing the vector $\hat{\zeta}$, one rearranges the indices such that the first asset has demonstrated the highest Sharpe ratio. This is to avoid the cumbersome notation of $\hat{\zeta}_{(1)}$, and we instead can just write $\hat{\zeta}_1$. We note this maximum condition can be written in the form $\mathbf{A}\hat{\zeta} \leq \mathbf{b}$ for $(k-1) \times k$ matrix $\mathbf{A}$ defined by

$$\mathbf{A} = \begin{bmatrix} -1 & 1 & 0 & \dots & 0 \\ -1 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -1 & 0 & 0 & \dots & 1 \end{bmatrix},$$

and where $\mathbf{b}$ is the $(k-1)$ -dimensional zero vector. Also note that we are interested in performing inference on ζ_1 , which we can express as $\boldsymbol{\eta}^\top \boldsymbol{\zeta}$ for $\boldsymbol{\eta} = \mathbf{e}_1$.

Under these conditions, if only $\hat{\zeta}$ were normally distributed, one could use the following theorem due to Lee *et al.*:

Theorem 2.1 (Lee *et al.*, Theorem 5.2 [15]). *Suppose $\mathbf{y} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. Define $\mathbf{c} = \boldsymbol{\Sigma}\boldsymbol{\eta}/\boldsymbol{\eta}^\top \boldsymbol{\Sigma}\boldsymbol{\eta}$, and $\mathbf{z} = \mathbf{y} - \mathbf{c}\boldsymbol{\eta}^\top \mathbf{y}$. Let $\Phi(x)$ be the CDF of a standard normal, and let $F(x; a, b, 0, 1)$ be the CDF of a standard normal truncated to $[a, b]$:*

$$F(x; a, b, 0, 1) =_{\text{def}} \frac{\Phi(x) - \Phi(a)}{\Phi(b) - \Phi(a)}.$$

Let $F(x; a, b, \mu, \sigma^2)$ be the CDF of a general truncated normal, defined by

$$F(x; a, b, \mu, \sigma^2) = F\left(\frac{x - \mu}{\sigma}; \frac{a - \mu}{\sigma}, \frac{b - \mu}{\sigma}, 0, 1\right).$$

Then, conditional on $\mathbf{A}\mathbf{y} \leq \mathbf{b}$, the random variable

$$F(\boldsymbol{\eta}^\top \mathbf{y}; \mathcal{V}^-, \mathcal{V}^+, \boldsymbol{\eta}^\top \boldsymbol{\mu}, \boldsymbol{\eta}^\top \boldsymbol{\Sigma} \boldsymbol{\eta})$$

is Uniform on $[0, 1]$, where $\mathcal{V}^-$ and $\mathcal{V}^+$ are given by

$$\begin{aligned} \mathcal{V}^- &= \max_{j: (\mathbf{A}\mathbf{c})_j < 0} \frac{\mathbf{b}_j - (\mathbf{A}\mathbf{z})_j}{(\mathbf{A}\mathbf{c})_j}, \\ \mathcal{V}^+ &= \min_{j: (\mathbf{A}\mathbf{c})_j > 0} \frac{\mathbf{b}_j - (\mathbf{A}\mathbf{z})_j}{(\mathbf{A}\mathbf{c})_j}. \end{aligned}$$

This theorem gives us a way to perform hypothesis tests, by comparing $F(\boldsymbol{\eta}^\top \mathbf{y}; \mathcal{V}^-, \mathcal{V}^+, \boldsymbol{\eta}^\top \boldsymbol{\mu}, \boldsymbol{\eta}^\top \boldsymbol{\Sigma} \boldsymbol{\eta})$ to some cutoff. It also suggests a procedure for computing confidence intervals on $\boldsymbol{\eta}^\top \boldsymbol{\mu}$, namely by univariate search for a value of $\boldsymbol{\eta}^\top \boldsymbol{\mu}$ such that $F(\boldsymbol{\eta}^\top \mathbf{y}; \mathcal{V}^-, \mathcal{V}^+, \boldsymbol{\eta}^\top \boldsymbol{\mu}, \boldsymbol{\eta}^\top \boldsymbol{\Sigma} \boldsymbol{\eta})$ is equal to some cutoff value.

In the following section we will show that the $\hat{\zeta}$ is *approximately* normally distributed. In the following section we will examine whether the normal approximation is good enough to use the procedure of Lee *et al.* for testing the signal-noise ratio of the asset with maximum Sharpe ratio.

2.1 Normal approximation of the distribution of Sharpe ratios

Here we derive the asymptotic distribution of Sharpe ratio, following Jobson and Korkie *inter alia*. [13, 17, 20, 22, 24] Consider the case of k possibly correlated returns streams, with each observation denoted by the k -vector $\mathbf{x}$. Let $\boldsymbol{\mu}$ be the k -vector of population means, and let $\boldsymbol{\alpha}_2$ be the k -vector of the uncentered second moments. Let $\boldsymbol{\zeta}$ be the vector of signal-noise ratios of the assets. Let r_0 be the ‘risk free rate’. We have

$$\zeta_i = \frac{\mu_i - r_0}{\sqrt{\alpha_{2,i} - \mu_i^2}}.$$

Consider the $2k$ vector of $\mathbf{x}$, ‘stacked’ with $\mathbf{x}$ squared elementwise, $[\mathbf{x}^\top, \mathbf{x}^2]^\top$. The expected value of this vector is $[\boldsymbol{\mu}^\top, \boldsymbol{\alpha}_2^\top]^\top$; let Ω be the variance of this vector, assuming it exists.

Given n observations of $\mathbf{x}$, consider the simple sample estimate

$$[\hat{\boldsymbol{\mu}}^\top, \hat{\boldsymbol{\alpha}}_2^\top]^\top =_{\text{df}} \frac{1}{n} \sum_i^n [\mathbf{x}^\top, \mathbf{x}^2]^\top.$$

Under the multivariate central limit theorem [32]

$$\sqrt{n} \left([\hat{\boldsymbol{\mu}}^\top, \hat{\boldsymbol{\alpha}}_2^\top]^\top - [\boldsymbol{\mu}^\top, \boldsymbol{\alpha}_2^\top]^\top \right) \rightsquigarrow \mathcal{N}(0, \Omega). \quad (1)$$

Let $\hat{\boldsymbol{\zeta}}$ be the sample Sharpe ratio computed from the estimates $\hat{\boldsymbol{\mu}}$ and $\hat{\boldsymbol{\alpha}}_2$: $\hat{\zeta}_i = (\hat{\mu}_i - r_0) / \sqrt{\hat{\alpha}_{2,i} - \hat{\mu}_i^2}$. By the multivariate delta method,

$$\sqrt{n} (\hat{\boldsymbol{\zeta}} - \boldsymbol{\zeta}) \rightsquigarrow \mathcal{N} \left(0, \left(\frac{d\boldsymbol{\zeta}}{d[\boldsymbol{\mu}^\top, \boldsymbol{\alpha}_2^\top]^\top} \right) \Omega \left(\frac{d\boldsymbol{\zeta}}{d[\boldsymbol{\mu}^\top, \boldsymbol{\alpha}_2^\top]^\top} \right)^\top \right). \quad (2)$$

Here the derivative takes the form of two $k \times k$ diagonal matrices pasted together side by side:

$$\begin{aligned} \frac{d\boldsymbol{\zeta}}{d[\boldsymbol{\mu}^\top, \boldsymbol{\alpha}_2^\top]^\top} &= \left[\text{Diag} \left(\frac{\boldsymbol{\alpha}_2 - \boldsymbol{\mu} r_0}{(\boldsymbol{\alpha}_2 - \boldsymbol{\mu}^2)^{3/2}} \right) \quad \text{Diag} \left(\frac{r_0 - \boldsymbol{\mu}}{2(\boldsymbol{\alpha}_2 - \boldsymbol{\mu}^2)^{3/2}} \right) \right], \\ &= \left[\text{Diag} \left(\frac{\boldsymbol{\sigma} + \boldsymbol{\mu} \boldsymbol{\zeta}}{\boldsymbol{\sigma}^2} \right) \quad \text{Diag} \left(\frac{-\boldsymbol{\zeta}}{2\boldsymbol{\sigma}^2} \right) \right]. \end{aligned} \quad (3)$$

where $\text{Diag}(\mathbf{z})$ is the matrix with vector $\mathbf{z}$ on its diagonal, and where the vector operations above are all performed elementwise, where we define the vector $\boldsymbol{\sigma} =_{\text{df}} (\boldsymbol{\alpha}_2 - \boldsymbol{\mu}^2)^{1/2}$, with powers taken elementwise.

In practice, the population values, $\boldsymbol{\mu}$, $\boldsymbol{\alpha}_2$, Ω are all unknown, and so the asymptotic variance has to be estimated, using the sample. This is impractical for large k , so instead one may wish to impose some distributional assumptions on $\mathbf{x}$.

Consider the case where $\mathbf{x}$ is drawn from a normal distribution with mean $\boldsymbol{\mu}$ and covariance Σ . Then, using Isserlis’ Theorem [12, 8], we have

$$\Omega = \begin{bmatrix} \Sigma & 2\Sigma \text{Diag}(\boldsymbol{\mu}) \\ 2\text{Diag}(\boldsymbol{\mu})\Sigma & 2\Sigma \odot \Sigma + 4\text{Diag}(\boldsymbol{\mu})\Sigma \text{Diag}(\boldsymbol{\mu}) \end{bmatrix}, \quad (4)$$

where $\odot$ denotes *Hadamard multiplication*.

Let $\mathbf{R}$ be the correlation matrix of the returns, defined as

$$\mathbf{R} =_{\text{df}} \text{Diag}(\boldsymbol{\sigma}^{-1}) \boldsymbol{\Sigma} \text{Diag}(\boldsymbol{\sigma}^{-1}), \quad (5)$$

where $\boldsymbol{\sigma}$ is the (positive) square root of the diagonal of $\boldsymbol{\Sigma}$. Then using the Ω given in Equation 4, Equation 1 becomes

$$\hat{\boldsymbol{\zeta}} \approx \mathcal{N}\left(\boldsymbol{\zeta}, \frac{1}{n} \left(\mathbf{R} + \frac{1}{2} \text{Diag}(\boldsymbol{\zeta}) (\mathbf{R} \odot \mathbf{R}) \text{Diag}(\boldsymbol{\zeta}) \right)\right). \quad (6)$$

(See the appendix.) Note how in the case of scalar Gaussian returns, this reduces to the well known standard error estimate of $\sqrt{\frac{1}{n} \left(1 + \frac{\zeta^2}{2} \right)}$. [17, 20, 3, 24] In practice the correlation matrix $\mathbf{R}$ and the vector of signal-noise ratios, $\boldsymbol{\zeta}$, have to be estimated and plugged in.

We claim that for the case of *elliptically distributed* $\mathbf{x}$, Equation 6 can be generalized to

$$\hat{\boldsymbol{\zeta}} \approx \mathcal{N}\left(\boldsymbol{\zeta}, \frac{1}{n} \left(\mathbf{R} + \frac{\kappa - 1}{4} \boldsymbol{\zeta} \boldsymbol{\zeta}^\top + \frac{\kappa}{2} \text{Diag}(\boldsymbol{\zeta}) (\mathbf{R} \odot \mathbf{R}) \text{Diag}(\boldsymbol{\zeta}) \right)\right), \quad (7)$$

where κ is the “kurtosis factor”, equal to one third the kurtosis of the marginals. [30] However, elliptically distributed returns have no skew, which makes them less than ideal for modeling returns series. Once again note how this equation reduces to the form of the standard error described by Mertens in the case of $k = 1$. [20]

Corollary 2.2 (to Theorem 2.1). *Let $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, with $\boldsymbol{\zeta} = \boldsymbol{\mu} \oslash \boldsymbol{\sigma}$, where $\boldsymbol{\sigma} = \text{diag}(\boldsymbol{\Sigma})$. Let $\mathbf{R}$ be the correlation matrix. Suppose you observe n independent observations of $\mathbf{x}$ then construct the Sharpe ratio, $\hat{\boldsymbol{\zeta}}$. Then, conditional on $A\hat{\boldsymbol{\zeta}} \leq \mathbf{b}$, the random variable*

$$u = F\left(\boldsymbol{\eta}^\top \hat{\boldsymbol{\zeta}}; \mathcal{V}^-, \mathcal{V}^+, \boldsymbol{\eta}^\top \boldsymbol{\zeta}, \boldsymbol{\eta}^\top \mathbf{Q} \boldsymbol{\eta}\right)$$

is Uniform on $[0, 1]$, where $\mathcal{V}^-$, $\mathcal{V}^+$ and $F(x; a, b, \mu, \sigma^2)$ are as in the theorem, and

$$\mathbf{Q} = \frac{1}{n} \left(\mathbf{R} + \frac{1}{2} \text{Diag}(\boldsymbol{\zeta}) (\mathbf{R} \odot \mathbf{R}) \text{Diag}(\boldsymbol{\zeta}) \right),$$

as given in Equation 6.

Note that the relationship between $\boldsymbol{\eta}^\top \hat{\boldsymbol{\zeta}}$ and $\mathcal{V}^-$ and $\mathcal{V}^+$ is such that u is unlikely to be strictly monotonic increasing with $\boldsymbol{\eta}^\top \hat{\boldsymbol{\zeta}}$, *ceterus paribus*. However, when $\boldsymbol{\eta}^\top \hat{\boldsymbol{\zeta}} \rightarrow \infty$, we expect $u \rightarrow 1$, and so to test the null hypothesis

$$H_0 : \boldsymbol{\eta}^\top \boldsymbol{\zeta} = c \quad \text{versus} \quad H_1 : \boldsymbol{\eta}^\top \boldsymbol{\zeta} > c,$$

one should reject at the α level when

$$F\left(\boldsymbol{\eta}^\top \hat{\boldsymbol{\zeta}}; \mathcal{V}^-, \mathcal{V}^+, c, \boldsymbol{\eta}^\top \mathbf{Q} \boldsymbol{\eta}\right) \geq 1 - \alpha.$$

As stated the procedure requires that one estimate $\mathbf{Q}$, which requires one to estimate $\mathbf{R}$ and $\boldsymbol{\zeta}$. However, computing the test statistic only requires access to

Q η . In the main inferential task considered here, that vector is the covariance of the asset with maximum Sharpe ratio against all the rest.

Note that Corollary 2.2 has uses beyond the stated problem of performing inference on the asset with the largest Sharpe ratio. For example, suppose you observe the Sharpe ratios of k assets, then select the asset with the largest *absolute* Sharpe ratio, choosing whether to hold it long or short depending on the sign of the Sharpe ratio. You wish to perform inference on your strategy. In this case, again reorder the assets such that the first asset has the highest absolute Sharpe ratio, but also flip the signs of the asset returns as necessary such that all assets have positive Sharpe ratio. Then proceed as in the usual case, but add to $\mathbf{A}$ and $\mathbf{b}$ the conditional restriction that all elements of $\hat{\zeta}$ are non-negative.

One wishes to also use the result for more general problems wherein one will hold a *portfolio* of assets, conditional on some observed properties of $\hat{\zeta}$. For example:

- Suppose you observe the Sharpe ratios of k assets, then select the top m by Sharpe ratio, then you choose to hold an some portfolio of those m assets. In this case set $\mathbf{A}$ and $\mathbf{b}$ to reflect the “ m choose $k - m$ ” relevant inequalities to condition on.
- Suppose you observe the Sharpe ratios of k assets, then select all assets with Sharpe ratio greater than some minimum value, ζ_* . Then you choose to hold some portfolio of all assets that pass the bar. In this case you need to modify $\mathbf{A}$ and $\mathbf{b}$ to condition on the passing assets having Sharpe ratios greater than ζ_* and the remaining assets having lower Sharpe ratio.

In these cases, the test vector η should reflect the chosen portfolio, but the signal-noise ratio of a portfolio is *not* the portfolio-weighted sum (or average) of the signal-noise ratios of the constituent assets. Indeed the signal-noise ratio of dollar-weighted portfolio ν is $\mu^\top \nu / \sqrt{\nu^\top \Sigma \nu}$. However, if ν is expressed in *volatility units*, then the signal-noise ratio is $\zeta^\top \nu / \sqrt{\nu^\top \mathbf{R} \nu}$. Thus assuming you can estimate volatility (and $\mathbf{R}$) without error², then one could transform a dollar denominated portfolio into a volatility denominated portfolio. From this one can perform inference using the test vector $\eta = \nu / \sqrt{\nu^\top \mathbf{R} \nu}$.

One could also use the procedure to test the hypothesis that the asset with maximum Sharpe ratio has higher signal-noise ratio than the *average* signal-noise ratio of all assets considered. This is the null commonly tested by the Reality Check and its variants. It may be of limited practical utility, however, since the selected asset may still have inferior signal-noise ratio.

3 Alternative Approaches

Before considering alternative approaches, it is worthwhile to step back and illustrate how they fundamentally differ from the conditional approach considered above. First, in Figure 1 we illustrate the observable flow chart of a practicing quantitative strategist. By some process the quant specifies a collection of strategies to test, then backtests them. We assume that the strategy with the

²Typically the error in a volatility estimate is less critical than error in the estimate of the mean. [5]

highest observed Sharpe ratio is then selected as a candidate for trading. Some kind of statistical test is performed on the whole collection of observed backtests. Depending on the outcome of that test, the selected strategy is either traded or not³.

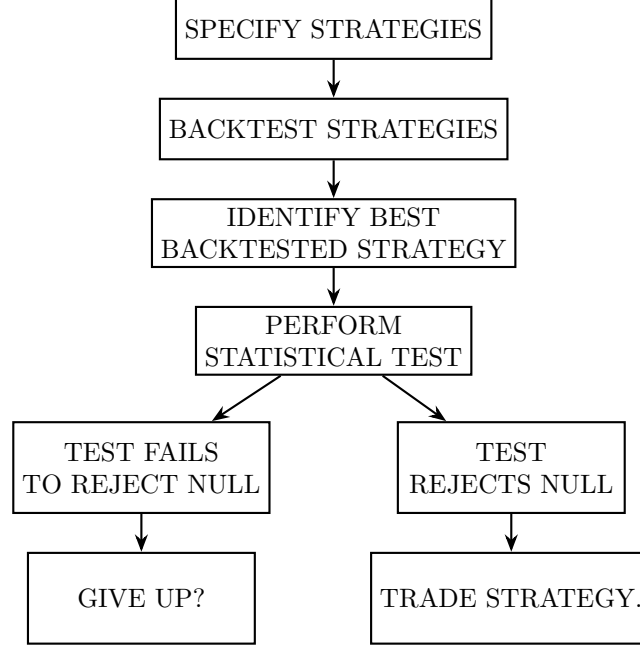


Figure 1: A quant strategist's statistical workflow is illustrated.

However, this flowchart illustrates only the outwardly observable parts of this workflow. In Figure 2 we include some of the latent states of this process and mark some of the unconditional probabilities of events. We treat the outcome of the strategy specification as potentially random, with some probability α_0 of yielding only “bad” strategies (those with signal-noise ratio below the threshold value), and probability $\alpha_1 = 1 - \alpha_0$ of yielding some “good” strategies. The quant then backtests all the strategies and selects the one with the best observed Sharpe ratio. There is some probability that this may be a good or bad strategy. The total unconditional probability that the selected strategy is bad is $\beta_0 + \beta_1$, while the probability of a good strategy is $\beta_2 = 1 - \beta_0 - \beta_1$. We then imagine the outcome of some statistical test, which may reject or fail to reject the null hypothesis. The probabilities of these events are illustrated in the figure and we denote their unconditional probabilities as γ_0 through γ_5 .

The point of this figure is to illustrate the difference between the conditional procedure and the procedures below, which test a different null hypothesis than the conditional procedure. The conditional procedure approximately controls the rate of type I errors which can be written as

$$\frac{\gamma_1 + \gamma_3}{\beta_0 + \beta_1}.$$

³Presumably if strategy does not pass the test, the quant starts all over again, or pursues a different career.

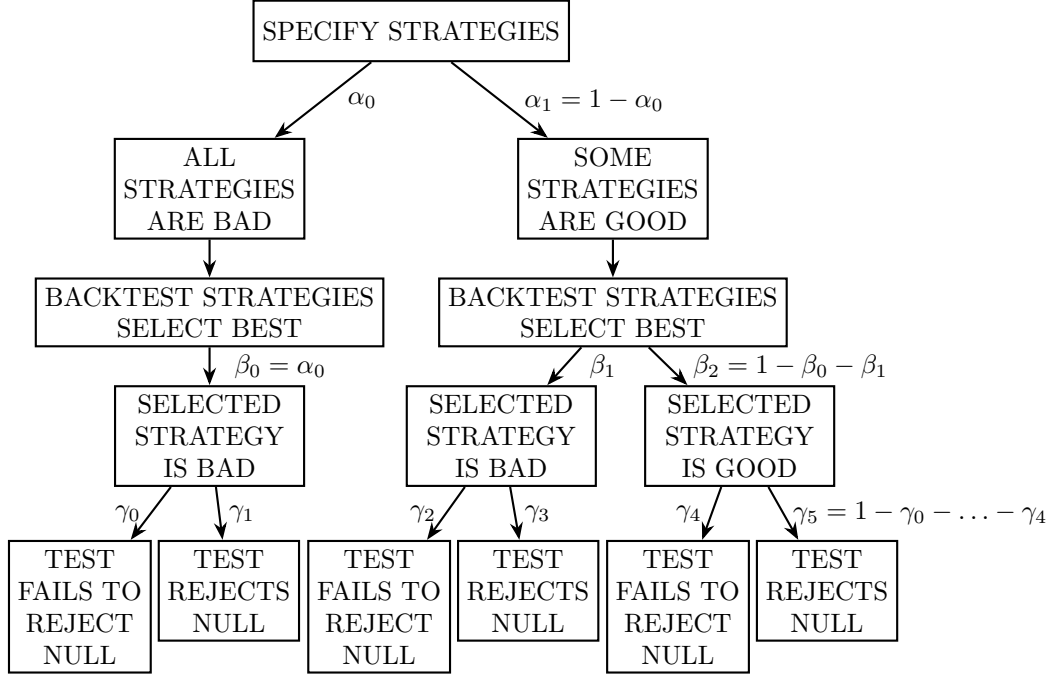


Figure 2: A quant strategist's statistical workflow with latent states is illustrated. The unconditional probabilities in each row sum to 1.0. The classical Bonferroni procedure defines the rate of type I errors as γ_1/β_0 , in contrast to the conditional test.

The power of the conditional procedure is the ratio γ_5/β_2 .

The Bonferroni procedure, and indeed all the procedures introduced in this section, only provide control over the type I rate γ_1/β_0 . The power of these procedures is

$$\frac{\gamma_3 + \gamma_5}{\beta_1 + \beta_2}.$$

There is considerable tension between the power and type I rate of this null versus that of the conditional procedure. Indeed in simulations we will find instances where $\gamma_3/\beta_1 \approx 1.0$ for some tests, although β_1 will be small for these cases.

Using this diagram we can also easily illustrate what [Storey](#) called the *positive False Discovery Rate* (pFDR). [29] The pFDR is the conditional probability of a false discovery conditional on the statistical test having rejected the null. In our diagram it is computed as

$$pFDR = \frac{\gamma_1 + \gamma_3}{\gamma_1 + \gamma_3 + \gamma_5}.$$

We would argue that pFDR is more important to control than the type I rate of a test. However, the pFDR depends strongly on the configuration of the population values, so it is hard to quantify it in a way that is relevant in all situations. Similarly it is hard to describe and test the null of the conditional procedure. The Bonferroni null, however, is easy to describe and test: simply set

the signal-noise ratio of every strategy to the null value. That is, the population ζ takes only one value under the Bonferroni null, but can have many different values under the conditional null. This will become more clear in the section on simulations under the null and alternative.

3.1 Bonferroni correction with simple correlation fix

The simple MHT approach to the problem is via a Bonferroni correction. [4] In its usual form, it assumes that the returns $\mathbf{x}$ are independent and normally distributed. In this case the marginals of $\hat{\zeta}$ are independent, and distributed as rescaled non-central t random variables. So to test the null hypothesis

$$H_0 : \forall_i \zeta_i = c \quad \text{versus} \quad H_1 : \exists_i \zeta_i > c, \quad (8)$$

compute the Sharpe ratios, $\hat{\zeta}$, then reject at the α level when $\sqrt{n} \max_i \hat{\zeta}_i$ exceeds $t_{1-\alpha/k}(\sqrt{nc}, n-1)$, the $1-\alpha/k$ quantile of the non-central t -distribution with $n-1$ degrees of freedom and non-centrality parameter $\sqrt{nc}$.

This simple test does not maintain nominal type I rate in the face of correlated assets. This can be demonstrated empirically, as we do in the sequel. One can get also get a theoretical hint of why this holds by considering the normal approximation of $\hat{\zeta}$ given in Equation 6, then appealing to Slepian's Lemma. Slepian's Lemma establishes that for a normally distributed Gaussian vector with fixed mean and variance, the maximum element is 'stochastically decreasing' as correlations increase. [28] Intuitively the number of true independent assets is decreasing as correlation increases.

Let us consider a simple model for the correlation matrix

$$\mathbf{R} = \rho(\mathbf{1}\mathbf{1}^\top) + (1-\rho)\mathbf{I}, \quad (9)$$

where $|\rho| \leq 1$. Now simplify Equation 6 to

$$\hat{\zeta} \approx \mathcal{N}\left(\zeta, \frac{1}{n}\mathbf{R}\right), \quad (10)$$

which is reasonable in the case of the small signal-noise ratios likely to be encountered in practice. Then under the null hypothesis that $\zeta = \zeta_0$, one observes

$$\mathbf{z} = \sqrt{n}\mathbf{R}^{-1/2}(\hat{\zeta} - \zeta_0) \approx \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (11)$$

where $\mathbf{R}^{-1/2}$ is the inverse of the (symmetric) square root of $\mathbf{R}$.

Under the assumed form for $\mathbf{R}$ given in Equation 9, it is simple to confirm that

$$\mathbf{R}^{-1/2} = \frac{1}{k} \left(\frac{1}{\sqrt{1-\rho+k\rho}} - \frac{1}{\sqrt{1-\rho}} \right) (\mathbf{1}\mathbf{1}^\top) + (1-\rho)^{-1/2}\mathbf{I}. \quad (12)$$

(This relation holds if we replace $\mathbf{1}\mathbf{1}^\top$ in $\mathbf{R}$ with $\mathbf{w}\mathbf{w}^\top$ where $\mathbf{w}$ is any vector whose elements are ± 1 . However in this case we will lose the order-preserving property.)

Now it is simple to confirm that in this case the transform induced by $\mathbf{R}^{-1/2}$ is "order-preserving." That is, if $\mathbf{a} = \mathbf{R}^{-1/2}\mathbf{b}$ and $b_i \leq b_j$ then $a_i \leq a_j$. As a consequence, if i is the maximal element of $(\hat{\zeta} - \zeta_0)$, then i is the maximal

element of $\mathbf{z}$. Let us assume, again, that by convention we have reordered the elements such that the first element of $\hat{\boldsymbol{\zeta}}$ is the maximum. Then to test the null hypothesis $\forall_i \zeta_i = c$, compute

$$z_1 = \frac{\sqrt{n}\hat{\zeta}_1}{\sqrt{1-\rho}} + \left(\frac{1}{\sqrt{1-\rho+k\rho}} - \frac{1}{\sqrt{1-\rho}} \right) \sqrt{n} \left(\frac{\mathbf{1}^\top \hat{\boldsymbol{\zeta}}}{k} - c \right), \quad (13)$$

and reject the null hypothesis when z_1 is bigger than $z_{1-\alpha/k}$, the $1-\alpha/k$ quantile of the normal distribution. In practice ρ must be estimated. This could be done by computing the correlations of the first asset against all others, then taking the average.

Note that the test statistic z_1 in Equation 13 depends on elements of $\hat{\boldsymbol{\zeta}}$ other than $\hat{\zeta}_1$. Indeed it depends on the average value among the $\hat{\zeta}_i$. This may not be desirable, as it would reject as one of the $\hat{\zeta}_i$ went to $-\infty$ for $i \neq 1$. Moreover, the statistic z_1 does not seem to be entirely “about” $\hat{\zeta}_1$, but is computed from all elements of $\hat{\boldsymbol{\zeta}}$. To rectify this, one is tempted to rotate the $\mathbf{z}$ from Equation 11 to be maximally aligned with $\mathbf{e}_1$. This is an area of continued research.

Note that we did not have to make the simplifying assumption that led from Equation 7 to Equation 10 given the form we assumed for $\mathbf{R}$. That is, assuming $\mathbf{R} = a_2 (\mathbf{1}\mathbf{1}^\top) + a_1 \mathbf{I}$, then under the null hypothesis that $\boldsymbol{\zeta} = \zeta_0 \mathbf{1}$, Equation 7 becomes

$$\hat{\boldsymbol{\zeta}} \approx \mathcal{N} \left(\boldsymbol{\zeta}, \frac{1}{n} (a'_2 (\mathbf{1}\mathbf{1}^\top) + a'_1 \mathbf{I}) \right), \quad (14)$$

for some constants a'_1, a'_2 which depend on a_1, a_2, κ and ζ_0 . One could then proceed as above, constructing a z_1 statistic.

Bonferroni correction for arbitrary correlation structure The Bonferroni correction outlined above is strictly only applicable to the rank-one correlation matrix, $\mathbf{R} = \rho (\mathbf{1}\mathbf{1}^\top) + (1-\rho)\mathbf{I}$. To apply the correction to any correlation matrix with positive entries, Slepian’s lemma allows us to appeal to a worst-case rank-one correlation matrix. [28, 35] Let $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{R})$ and $\mathbf{y} \sim \mathcal{N}(\mathbf{0}, \rho (\mathbf{1}\mathbf{1}^\top) + (1-\rho)\mathbf{I})$, for $\rho \geq 0$ where $R_{i,j} \geq \rho$ for all $i \neq j$. By Slepian’s lemma, $\Pr \{\max_i x_i \geq t\} \leq \Pr \{\max_i y_i \geq t\}$.

Then assume the correlation of returns is $\mathbf{R}$, where $R_{i,j} \geq \rho$ for $i \neq j$ for some $\rho \geq 0$; to test the null hypothesis $\forall_i \zeta_i = c$, compute z_1 as in Equation 13, plugging in ρ , and reject the null hypothesis when z_1 is bigger than $z_{1-\alpha/k}$, the $1-\alpha/k$ quantile of the normal distribution. This procedure has (approximate) type I rate *no greater* than α .

3.2 Testing against one-sided alternatives

Another obvious approach to the problem is to appeal to the normal approximation of Equation 6 or Equation 7, then use well known techniques in testing of a multivariate normal against a one-sided alternative. [27, 31] That is, the usual multivariate procedure to test the null hypothesis $H_0 : \forall_i \zeta_i = c$ under a normal approximation would be via Hotelling’s T^2 test. [1, 25] However we are not interested in the case where some of the ζ_i are less than c .

One-sided tests will not scale well to the case of large k , except perhaps under simple models for correlation. Consider testing the following null

$$H_0 : \forall_i \zeta_i = \zeta_0 \quad \text{versus} \quad H_1 : \forall_i \zeta_i \geq \zeta_0 \text{ and } \exists_j \zeta_j > \zeta_0,$$

subject to the rank one correlation structure of Equation 9 with $\rho \geq 0$. Again assume the approximation of Equation 10,

$$\hat{\zeta} \approx \mathcal{N}\left(\zeta, \frac{1}{n}\mathbf{R}\right).$$

Letting $\hat{\xi} = \mathbf{R}^{-1/2}\hat{\zeta}$ and $\xi = \mathbf{R}^{-1/2}\zeta$, the normal approximation can be rewritten as

$$\sqrt{n}\hat{\xi} \approx \mathcal{N}(\sqrt{n}\xi, \mathbf{I}).$$

The inverse square root of $\mathbf{R}$, given in Equation 12, is order-preserving. Moreover,

$$\mathbf{R}^{-1/2}\mathbf{1} = c\mathbf{1},$$

for $c = (1 + (k-1)\rho)^{-1/2} > 0$. From the order-preserving nature of $\mathbf{R}^{-1/2}$, the null and alternative hypotheses can be expressed as

$$H_0 : \forall_i \xi_i = c\zeta_0 \quad \text{versus} \quad H_1 : \forall_i \xi_i \geq c\zeta_0 \text{ and } \exists_j \xi_j > c\zeta_0.$$

We can then appeal to the simple chi-bar square test. [27, 34] First transform the vector of Sharpe ratios to $\hat{\xi}$ via

$$\hat{\xi} = \mathbf{R}^{-1/2}\hat{\zeta} = c\bar{\zeta}\mathbf{1} + (1 - \rho)^{-1/2}(\hat{\zeta} - \bar{\zeta}\mathbf{1}), \quad (15)$$

where $\bar{\zeta}$ is the average of the sample Sharpe ratios. Then compute

$$\bar{x}^2 = n \sum_i \left(\hat{\xi}_i - c\zeta_0 \right)_+^2, \quad (16)$$

where y_+ is the positive part of y , *i.e.*, $y_+ = y$ if $y > 0$ and zero otherwise. Then compute the CDF of the corresponding chi-bar square distribution as

$$Q = \sum_{i=0}^k w_i F_{\chi^2}(\bar{x}^2; i), \quad (17)$$

where $F_{\chi^2}(x; i)$ is the cumulative distribution of the χ^2 distribution with i degrees of freedom, and w_i are the chi-bar square weights. In this case they are defined as

$$w_i = \binom{k}{i} 2^{-k}.$$

Reject the null hypothesis at the α level if $1 - Q \leq \alpha$.

Note that, as with the Bonferroni correction, the test statistic $\bar{x}^2$ is computed on all elements of $\hat{\zeta}$, and thus the decision to reject the null may not be “about” $\hat{\zeta}_1$. In testing we will see that the one-sided test is highly susceptible to distribution of the ζ , moreso than the Bonferroni correction.

We note that under this setup it is also easy to use Follman’s test, which is a very simple procedure with increased power against one-sided alternatives. [6] Here we would compute

$$g^2 = nkc^2 (\bar{\zeta} - \zeta_0)^2 + \frac{n}{1 - \rho} \sum_i \left(\hat{\zeta}_i - \bar{\zeta} \right)^2,$$

and reject at the α level if both $1 - F_{\chi^2}(g^2; k) \leq 2\alpha$ and $\bar{\zeta} > \zeta_0$.

3.3 Hansen’s log log Corrections

One failing of many of the approaches considered above is the problem of irrelevant alternatives. That is, instead of testing under the null of equality, (8) above, we should test the following

$$H_0 : \forall_i \zeta_i \leq c \quad \text{versus} \quad H_1 : \exists_i \zeta_i > c.$$

Testing such a composite null hypothesis is typically via a *non-similar* test, *i.e.*, one which has a type I rate no greater than the nominal rate for all ζ in the null, and which achieves that nominal rate for some ζ under the null. Such tests achieve the nominal rate under the *least favorable configuration* (LFC), which in our case is the null of equality, or the problem of (8). [27]

Hansen describes a procedure which avoids this problem. The idea is elegant, and ultimately simple to implement. [10, 9] In the terms of the problem we consider, it amounts to assuming that the null takes the form

$$H_0 : \forall_i \zeta_i \leq c \text{ and } |\zeta_i - \hat{\zeta}_i| \leq g_n \quad \text{versus} \quad H_1 : \exists_i \zeta_i > c,$$

for some g_n . Note this seems odd since the sample Sharpe ratio appears in the null hypothesis to be tested. Hansen describes how such a test can be performed while maintaining a maximum type I rate asymptotically, and achieving higher power.

Hansen applied this correction to the chi-bar-square statistic (*cf.* Equation 16), and later to a Studentized version of White’s Reality Check statistic, which is rather like the corrected Bonferroni statistic computed in Equation 13. [10, 9] Applying Hansen’s correction to our problem is simple: compute $\hat{\xi}$ as in Equation 15. Let $\tilde{k}$ be the number of elements of $\hat{\xi}$ greater than $c\zeta_0 - \sqrt{(2 \log \log n)/n}$, where $c = (1 + (k-1)\rho)^{-1/2}$. If $\tilde{k} = 0$ fail to reject. Otherwise compute the chi-bar-square statistic $\bar{x}^2$ as in Equation 16 and reject if

$$\sum_{i=0}^{\tilde{k}} \binom{\tilde{k}}{i} 2^{-\tilde{k}} F_{\chi^2}(\bar{x}^2; i) \geq 1 - \alpha.$$

We will refer to this as “Hansen’s chi-bar-square.” It is the chi-bar-square test considered above, but with reduced degrees of freedom *which depend on the observed*.

The same correction is easily applied to the Bonferroni maximum test: again, compute $\hat{\xi}$ and $\tilde{k}$. If $\tilde{k} = 0$ fail to reject. Otherwise reject at the α level if

$$\max_i \hat{\xi}_i - c\zeta_0 \geq z_{1-\alpha/\tilde{k}}.$$

We will refer to this as “Hansen’s SPA,” although it does not use the bootstrap procedure to estimate the standard error as described by Hansen, it is similar in every other regard. [9]

3.4 Subspace approximation

Another potential approach to the problem which may be useful in the case where returns are highly correlated, as one expects when returns are from back-tested quantitative trading strategies, is via a subspace approximation. First

we assume that the $n \times k$ matrix of returns, $\mathbf{X}$ can be approximated by a m -dimensional subspace

$$\mathbf{X} \approx \mathbf{Y}\mathbf{W},$$

where $\mathbf{Y}$ is a $n \times m$ matrix of ‘latent’ returns, and $\mathbf{W}$ is a $m \times k$ ‘loading’ matrix.

Now the column of $\mathbf{X}$ with maximal $\hat{\zeta}$ has Sharpe ratio that is smaller than

$$\hat{\zeta}_* =_{\text{df}} \max_{\boldsymbol{\nu}} \frac{\hat{\boldsymbol{\mu}}^\top \boldsymbol{\nu}}{\sqrt{\boldsymbol{\nu}^\top \hat{\boldsymbol{\Sigma}} \boldsymbol{\nu}}},$$

where $\hat{\boldsymbol{\mu}}$ is the m -vector of the (sample) means of columns of $\mathbf{Y}$ and $\hat{\boldsymbol{\Sigma}}$ is the sample covariance matrix. This maximum takes value

$$\hat{\zeta}_* = \sqrt{\hat{\boldsymbol{\mu}}^\top \hat{\boldsymbol{\Sigma}}^{-1} \hat{\boldsymbol{\mu}}},$$

which is, up to scaling, Hotelling’s T^2 statistic.

Under the null hypothesis that the rows of $\mathbf{Y}$ are independent draws from a Gaussian random variable with zero mean, then

$$\frac{(n-m)\hat{\zeta}_*^2}{m(n-1)}$$

follows an F distribution with m and $n-m$ degrees of freedom. Under the alternative it follows a non-central F distribution. [1, 25] Via this upper bound $\hat{\zeta}_1 \leq \hat{\zeta}_*$, one can then perform tests on the null hypothesis $\forall_i \zeta_i = 0$.

However, this approach requires that one estimate m , the dimensionality of the latent subspace. Moreover, the subspace approximation may not be very good. It would seem that to get near equality of $\hat{\zeta}_1$ and $\hat{\zeta}_*$, the columns of $\mathbf{X}$ would have to contain both positive and negative exposure to the columns of $\mathbf{Y}$. This in turn should result in mixed correlation of asset returns, which we may not observe in practice. Finally, empirical testing indicates this approach requires further development. [23]

4 Empirical Results

4.1 Simulations under the null

4.1.1 Gaussian returns, infeasible estimator

First we seek to establish if, and under what circumstances, the normal approximation of Equation 6 is sufficiently accurate to give nominal coverage under the conditional estimation procedure. First we test a single case of $k = 100$ using a correlation matrix that is $\rho = 0.7$ on the off-diagonals: $\mathbf{R} = 0.7(\mathbf{1}\mathbf{1}^\top) + 0.3\mathbf{I}$. We generate Gaussian returns over 1260 days, approximately 5 years worth for equity returns. We let $\boldsymbol{\zeta}$ range uniformly from $-0.1\text{day}^{-1/2}$ to $0.1\text{day}^{-1/2}$. We compute the Sharpe ratios of each asset’s simulated returns, find the asset with maximum Sharpe ratio, then compute a p-value using Theorem 2.1. Since we wish to assess the accuracy of the normal approximation, we use the actual population value of $\mathbf{R}$, and the $\boldsymbol{\zeta}$ to compute the covariance via Equation 6. This is not, of course, how the test would be applied in practice since $\mathbf{R}$ and $\boldsymbol{\zeta}$ have

to be estimated. Moreover, we wish to check coverage of the procedure under the null, so we use the actual $\boldsymbol{\eta}^\top \boldsymbol{\zeta}$ in our test.

We repeat this experiment 10^5 times and collect the resultant putative p-values. We would like to Q-Q or P-P plot these p-values, as evidence that they are near uniform, but the large sample size presents some challenges. Instead, we choose some selected small q (like 0.05 or 0.01), and compute the proportion of our p-values $\leq q$. We then subtract q . This value, call it Δ , should be near zero. We plot Δ against q , with errorbars around the x axis reflecting the area where we would expect the points to fall roughly 95% of the time. Those errorbars are computed via the Binomial law, but do not always have width of exactly 95% because of the finite sample size. However, the plot suggests that the p-values are indeed uniformly distributed, and that the procedure has near nominal type I rate when selecting a cutoff near the selected q .

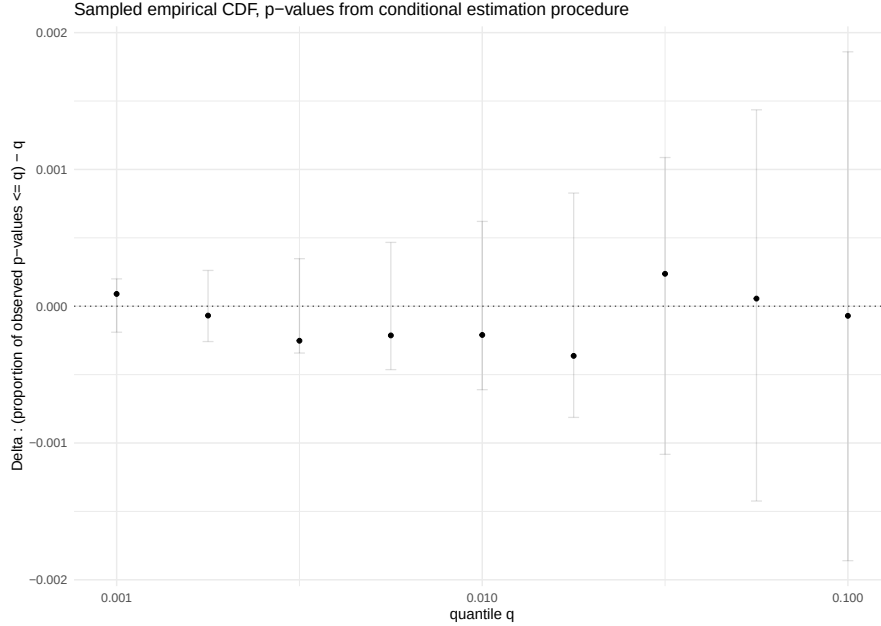


Figure 3: The computed p-values from the conditional estimation procedure over $1e+05$ simulations are compared to a uniform law. Given the large sample size, a Q-Q plot is visually hard to interpret. Instead, for selected q , we compute the proportion of our putative p-values less than q . We then compute $\Delta = (\text{Prop} \leq q) - q$ against q . Using the binomial law, we plot approximate error bars around the x axis that indicate where the points should fall with approximately 95% confidence. Simulations use the exact $\mathbf{R}$ and $\boldsymbol{\zeta}$ to compute the covariance matrix. The plot supports uniformity of the putative p-values.

4.1.2 Gaussian returns, feasible estimator

While these experiments suggest the normal approximation leads to nearly uniform p-values under the null, they use the (unknown) population values of R and ζ to compute the variance-covariance of $\hat{\zeta}$. So we repeat the experiments, but plug in the usual sample estimate of covariance and the vector of Sharpe ratios into Equation 6 to estimate the covariance matrix of $\hat{\zeta}$. Other than this change, we repeat the previous experiment, performing 10^5 simulations, setting $k = 100$, $R = 0.7(\mathbf{1}\mathbf{1}^\top) + 0.3I$. $n = 1260\text{day}$, *etc.* In Figure 4 we present a sampled CDF plot of the log p values, as above. Again the simulations are consistent with the procedure having nominal coverage. This is not surprising, because, as noted above, the statistical test only requires us to estimate the standard error of the Sharpe ratio of the asset with maximum Sharpe ratio, and so does not greatly rely on the k^2 elements of the estimate of R .

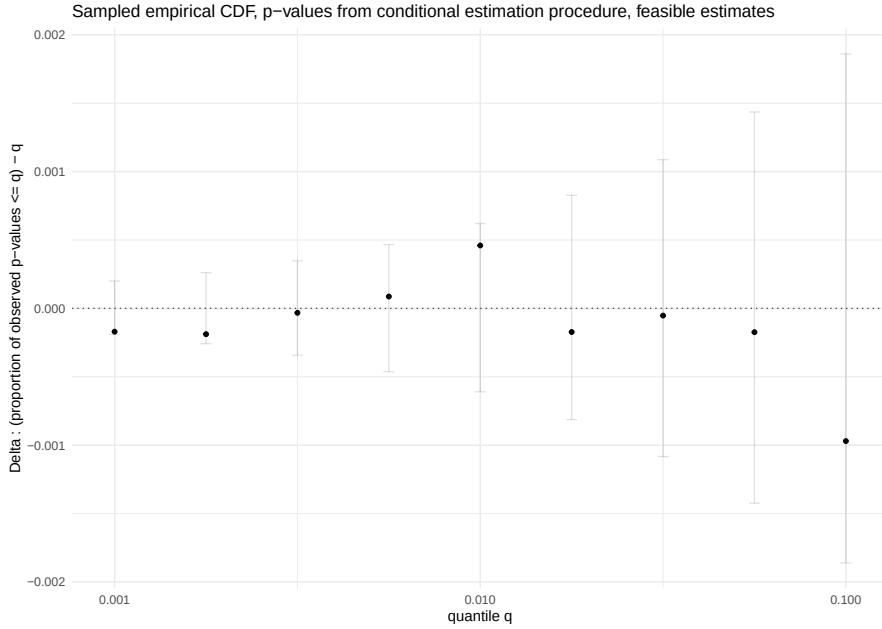


Figure 4: The computed p-values from the conditional estimation procedure using sample estimates of the covariance matrix, R , and the vector of signal-noise ratios, ζ are used in this sampled CDF plot. P values are computed over $1e+05$ simulations. For selected q , we compute the proportion of our putative p-values less than q . We then compute $\Delta = (\text{Prop} \leq q) - q$ against q . Using the binomial law, we plot approximate error bars around the x axis that indicate where the points should fall with approximately 95% confidence. Simulations use the exact R and ζ to compute the covariance matrix. The plot supports uniformity of the putative p-values.

4.1.3 Gaussian returns, feasible estimator, sensitivity

This kind of “proof by eyeball” is somewhat unsatisfying, and does not scale up to the task of finding where the approximation is accurate. To measure the uniformity of our p-values, we generate some via simulations as described above, then compute the Kolmogorov-Smirnov statistic against a uniform distribution. [19] You can think of the K-S statistic as the maximum absolute deviance of a point away from the $y = x$ line in a P-P plot like Figure 3.

So we repeat the previous experiments, using a feasible estimator of the covariance matrix of $\hat{\zeta}$. Again we draw returns from a Gaussian distribution. We let n vary from 63 to 500; we let k vary from 50 to 200; we let ρ vary from 0 to 0.8 where we take $R = \rho (\mathbf{1}\mathbf{1}^\top) + (1 - \rho) I$; we take ζ to be a uniform sequence from 0 to $0.1\text{day}^{-1/2}$. For each setting of the parameters in the Cartesian product we perform 10^5 simulations, computing p values from the feasible estimator.

In Figure 5 we plot those K-S statistics against n , with different facets for ρ . All else equal, we expect the approximation to be worse, and thus the K-S statistics to be higher, for smaller n and larger k . This pattern is somewhat visible in the plots, although large ρ seems to reduce the number of ‘pseudo-assets’ in that relationship, and the number of observations, n , appears to have greater impact than the number of assets k . However, with the given limited evidence, we cannot claim to have definitively established where our procedure breaks down, but warn users that the $k \gg n$ cases are likely to be problematic in the sense that nominal type I rates may not be maintained.

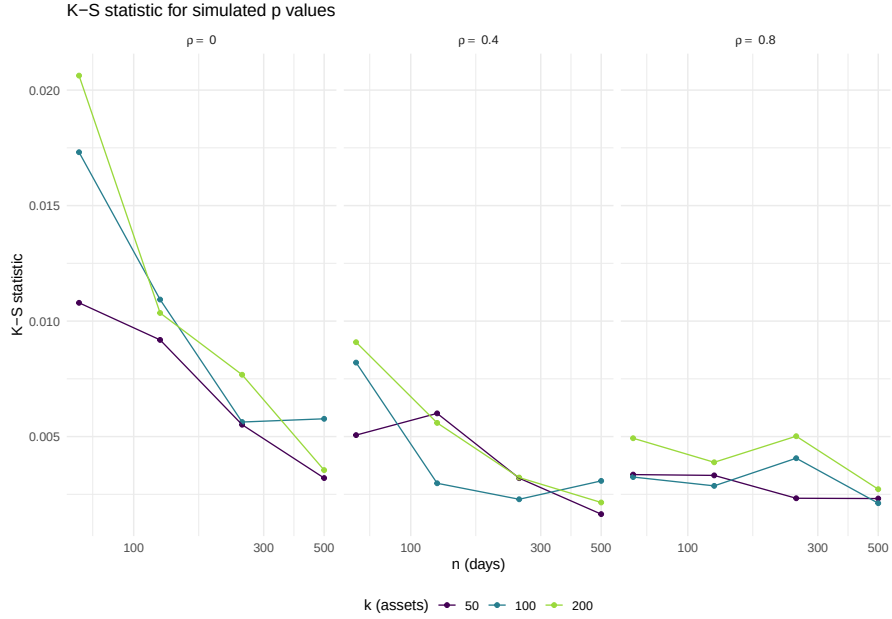


Figure 5: Kolmogorov-Smirnov statistics summarizing uniformity of the test statistic u are plotted versus n with facets for ρ . Broadly we see that the test statistic is less uniform in the regime where n is small, but this is muted by a large positive correlation ρ .

4.1.4 t returns, feasible estimator

The simulations above were carried out assuming Gaussian returns, and using Equation 6 to compute the covariance matrix of $\hat{\zeta}$. Gaussian returns are not a good model for real asset returns, so we repeat those simulations with returns drawn from a multivariate t -distribution with 5 degrees of freedom. [16, 14] Again we perform 10^4 simulations with $k = 100$, $R = 0.7(\mathbf{1}\mathbf{1}^\top) + 0.3\mathbf{I}$, $n = 1260$ day, *etc.* We perform inference twice, once using Equation 6, and once using Equation 7 where we have estimated the kurtosis factor, κ , by taking the median of the sample marginal kurtosis of the assets. In Figure 6 we present the subsampled empirical CDF plots on the log-transformed p-values under the two methods of estimating the covariance of $\hat{\zeta}$. There is little difference in the performance of the two sets of simulations, though without the correction, the procedure is slightly anti-conservative, while with the correction it is slightly conservative. We remain cautiously optimistic that for large n , one need not correct Equation 6 to account for non-normal returns.

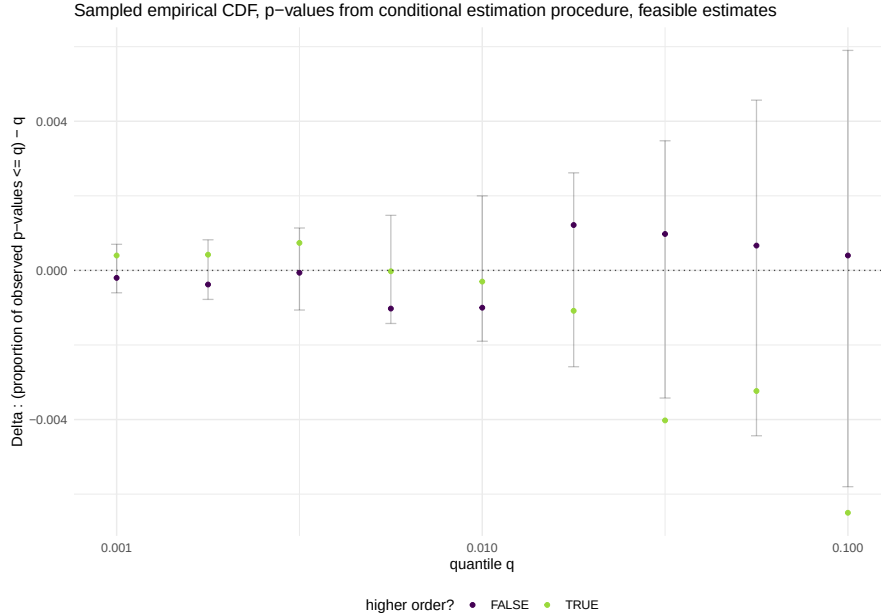


Figure 6: The computed p-values from the conditional estimation procedure over 10000 simulations are plotted using our sampled CDF procedure. Returns are drawn from a $t(5)$ distribution. Sample estimates are used to construct the variance-covariance matrix of $\hat{\zeta}$. The experiments are performed twice: first assuming that returns are Gaussian; then assuming returns are elliptical with unknown kurtosis factor that we estimate from the sample. Both procedures give near-uniform p values. Without correcting for higher order moments, the procedure has higher than nominal type I rate; with the correction it has slightly smaller than the nominal rate.

4.2 Simulations under the alternative

We wish to test the power of the method under the alternative hypothesis. However, it is hard to state exactly what constitutes *the* alternative. One interpretation is that we condition on $\zeta_1 > 0$, where again the indexing is such that $\hat{\zeta}_1$ was the maximum over k assets; then we estimate the probability of (correctly) rejecting $\zeta_1 = 0$ versus ζ_1 . However, we suspect that the power, as described in this way, would depend on the distribution of values of ζ , as well as the correlation of returns among the tested strategies.

We will consider following alternatives: one where all k elements of ζ are equal (“all-equal”), and three others where m of k elements of ζ are equal to some positive value, and the remaining $k - m$ are negative that value. For $m = 1$, we call this the “one-good” alternative; for $m = 2$, the “two-good” alternative; for $m = k/2$, the “half-good” alternative. Later we also consider the cases where the elements of ζ are uniformly distributed across some finite set of values (“uniform”) or distributed like the density of a scaled and shifted binomial distribution (“bell-shaped”).

We compare the power of the conditional estimation procedure to that of a simple MHT correction, and the one-sided test. In our experiments, we draw returns from a Gaussian distribution with diagonal covariance. Under this assumption, one can use the distribution of the t statistic to perform inference on the signal-noise ratio. [22, 24] We then use the Bonferroni correction to account for the multiple tests performed. We also perform the one-sided test based on the chi-bar square statistic, Follman’s test, and the chi-bar and MHT tests with Hansen’s log log adjustment.

Note that in the all-equal case, since every asset has the same signal-noise ratio, whichever we select will have the same signal-noise ratio, and the Bonferroni-corrected test should have the same power as the t -test for a single asset. The conditional estimation procedure, however, may suffer in this case as we may condition on a $\hat{\zeta}_1$ that is very close to being non-optimal, resulting in a small test statistic for which we do not reject. On the other hand, for the one-good case, as the $k - 1$ assets may have considerably negative signal-noise ratio, they are unlikely to exhibit the largest Sharpe ratio, and so the MHT is merely testing a single asset, but at the α/k level instead of the α level, resulting in lower power. The chi-bar square test and Follman’s test are also unlikely to reject. The conditional estimation procedure, however, should not suffer under this alternative. We also expect the log log adjustment to have little effect when the Sharpe ratios are all nearly equal, and more effect when they are different.

Our suspicions are born out by the simulations. We perform simulations under all-equal, half-good, two-good and one-good configurations, letting the ‘good’ signal-noise ratio vary from 0 to $0.212\text{day}^{-1/2}$, which corresponds to an ‘annualized’ signal-noise ratio of around $3.4\text{yr}^{-1/2}$. We draw Gaussian returns with diagonal covariance for 100 assets, with $n = 1008$. For each setting we perform 10^4 simulations then compute the empirical rejection rate of the test at the 0.05 level, conditional on the signal-noise ratio of the *selected* asset, which is to say the one with the largest Sharpe ratio. Note that in some simulations the largest Sharpe ratio is observed for an asset with a negative signal-noise ratio. We hope our tests to have lower power when this occurs.

In Figure 7, we plot the power of Follman’s test, Hansen’s chi-bar square and MHT (“SPA”) tests, and the conditional estimation procedure versus the signal-

noise ratio of the selected asset. We present facet columns for the configurations of ζ , *viz.* all-equal, half-good, two-good and one-good. A horizontal line at 0.05 gives the nominal rate under the null, which occurs as $x = 0$ in these plots.

As expected from the above explanation, the chi-bar square (with Hansen's correction) has the highest power for the all-equal alternative, followed by Follman's test, then the MHT, then the conditional estimation test. These relationships are nearly exactly reversed for the one-good case: Follman's test has zero power against the one-good alternative, while the conditional procedure has the highest power. Each test considered here shows similar performance in the half-good as in the all-equal alternative, with the exception of Follman's test, which achieves a maximum power of $1/2$ in the half-good case, as is to be expected since this is the probability that $\zeta > 0$ in the half-good case.

In Figure 8, we plot the power of the MHT and chi-bar square procedures, with and without Hansen's log-log corrections. The correction has essentially no effect in the all-equal and half-good cases, but increases power considerably for the chi-bar square test in the two-good and one-good alternatives, and increases power modestly for the MHT test under those same alternatives. Thus we mostly consider using the log-log correction when possible. We note that this correction may result in slightly higher than nominal type I rate, as evidenced later in Figure 15.

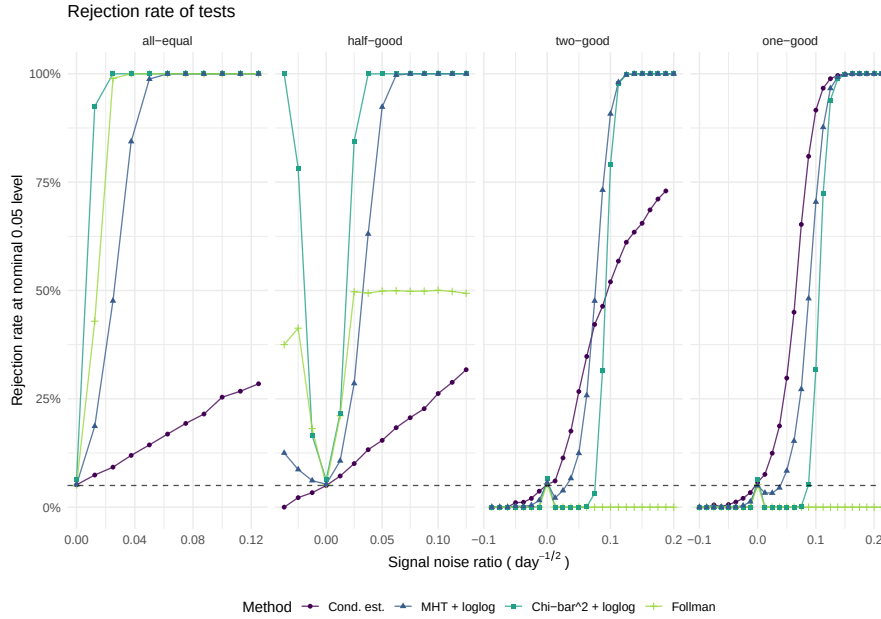


Figure 7: The empirical power of the conditional estimation test, Follman's test, and the chi-bar-square and MHT tests with Hansen's loglog correction are shown versus the signal-noise ratio of the asset with maximum Sharpe ratio under different arrangements of the vector ζ . For the one- and two-good cases, Follman's test has essentially zero power under the alternative.

The power of the conditional estimation procedure for the all-equal and half-good cases is rather disappointing. For the case where all assets have a

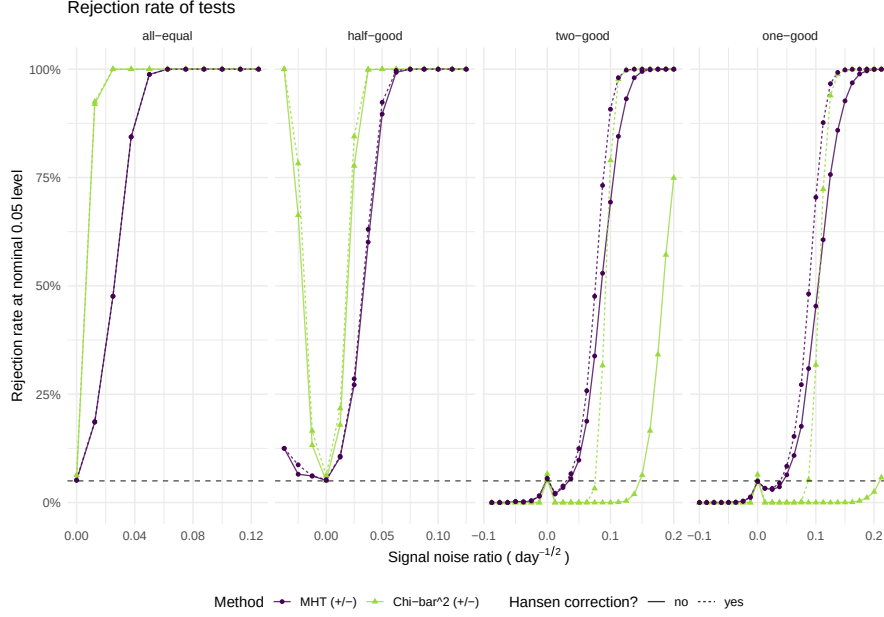


Figure 8: The empirical power of the MHT corrected test, the chi-bar-square test, with and without Hansen’s log log correction, are shown versus the signal-noise ratio of the asset with maximum Sharpe ratio under different arrangements of the vector ζ .

signal-noise ratio of $3.4\text{yr}^{-1/2}$, which should be considered very very large, the test has a power of only around a half. The test suffers from low power because we are conditioning on “ $\hat{\zeta}_1$ is the largest Sharpe ratio”, but we should actually like to condition on “the asset with the largest Sharpe ratio, whichever one that is.” That is, there is no recognition here that if another asset had exhibited higher Sharpe ratio, we would have selected that one instead. It is not clear if that distinction can be meaningfully integrated into this testing procedure.

However, we caution that the apparent low power of the conditional inference test in comparison to tests based on MHT is that the latter are rejecting a different null, and can actually have much higher type I rate under the conditional null. Among these tested alternatives, this effects is most visible in the half-good facet, where the MHT correction and one-sided tests have greater than 0.05 rejection rate for *negative* signal-noise ratio. Some clarification is required here. We have performed 10^4 simulations for each setting of the ‘good’ signal-noise ratio; in some number of them for the half-good case, an asset with negative signal-noise ratio exhibits the maximum Sharpe ratio. We are plotting the rejection rate for the test in this case. But note that the null hypothesis that MHT and the one-sided test are testing *is* violated in this case, because half the assets have positive signal-noise ratio, and the alternative procedures test the null that all assets have zero or lower signal-noise ratio. Tests based on the MHT do not maintain the nominal type I rate under the conditional null. While the probability of selecting a ‘bad’ asset instead of a ‘good’ one is going to be very low, especially when there is a large gap between the signal-noise

ratios of the good and the bad assets, it is troubling that when this occurs, tests based on the MHT can fail with high probability.

On the other hand, while the conditional estimation procedure exhibits lower power than the other tests (except in the one-good case), it appears to have monotonic rejection probability with respect to the signal-noise ratio of the *selected* asset in all configurations tested. That is, in the half-good case, it has low rejection probability in the odd simulations where a ‘bad’ asset is selected because it is actually designed for testing the conditional null. This pattern is repeated in the following set of simulations under the alternative.

It is unlikely that the four distributions of ζ considered above will be encountered in real world applications; rather we anticipate that the elements of ζ will have a distribution which is bell-shaped or even flat. We simulate these cases, estimating the power of the various tests as a function of the signal-noise ratio of the selected asset.

We set $k = 105$, and $n = 1008$. The elements of ζ are set to fixed values shown in Figure 9. Namely for the uniform case, the ζ takes uniform values from $-0.07\text{day}^{-1/2}$ to $0.07\text{day}^{-1/2}$, or in annualized units $[-1.1, 1.1]\text{day}^{-1/2}$. Under the bell-shaped simulations, the ζ takes the same range but with more mass around zero.

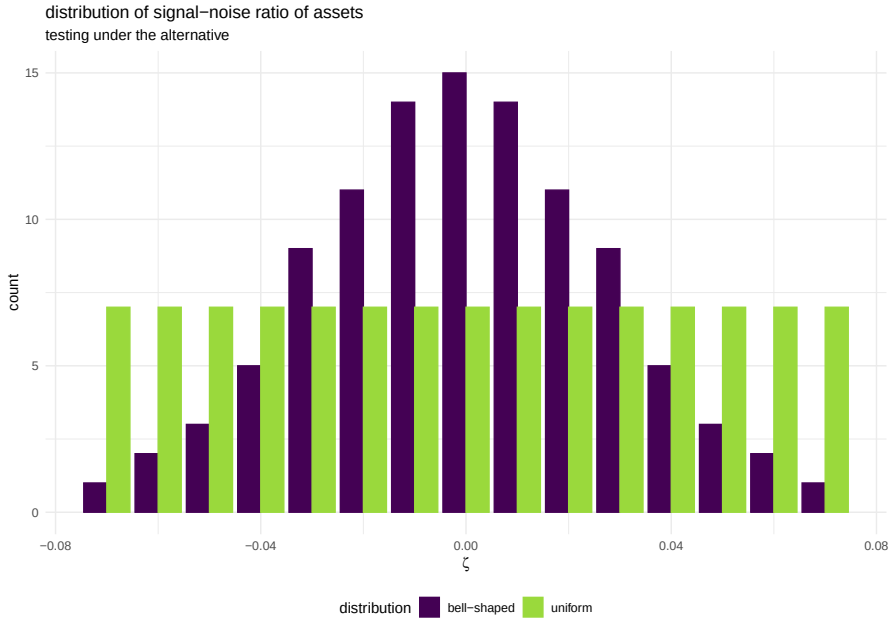


Figure 9: The counts of the values of ζ are shown for the uniform and bell-shaped simulations under the alternative. The signal-noise ratio is given in daily units, $\text{day}^{-1/2}$.

In Figure 10 we plot the empirical rejection rates for the various tests under these two distributions of ζ . The rejection rate is plotted against the signal-noise ratio of the selected strategy, ζ_1 . We note that in some unlucky cases ζ_1 is small or even negative even though the maximum element of ζ is significantly positive. As seen above, the conditional procedure maintains the near nominal

type I rate of 0.05 when $\zeta_1 = 0$, with modest power when $\zeta_1 > 0$. The other tests exhibit much higher rejection rates, even when $\zeta_1 \leq 0$. This is not unexpected, as the null *is* violated in this case and rejecting it is the right decision for tests based on the MHT null. We reiterate that tests based on the MHT can, in some minority of cases, correctly reject the null, but the selected strategy has unacceptable signal-noise ratio. In Figure 11 we plot the empirical probability under our simulations that the asset with maximum Sharpe ratio has a given signal-noise ratio. For both the uniform and bell-shaped ζ , the probability that $\zeta_1 \leq 0$ appears to be less than 10%, with smaller chance under the uniform distribution. Thus the likelihood of making this kind of conditional type I error when using MHT tests is perhaps low, it is not zero. However, if a conditional type I error rate must be maintained, the MHT tests should be considered unacceptable.

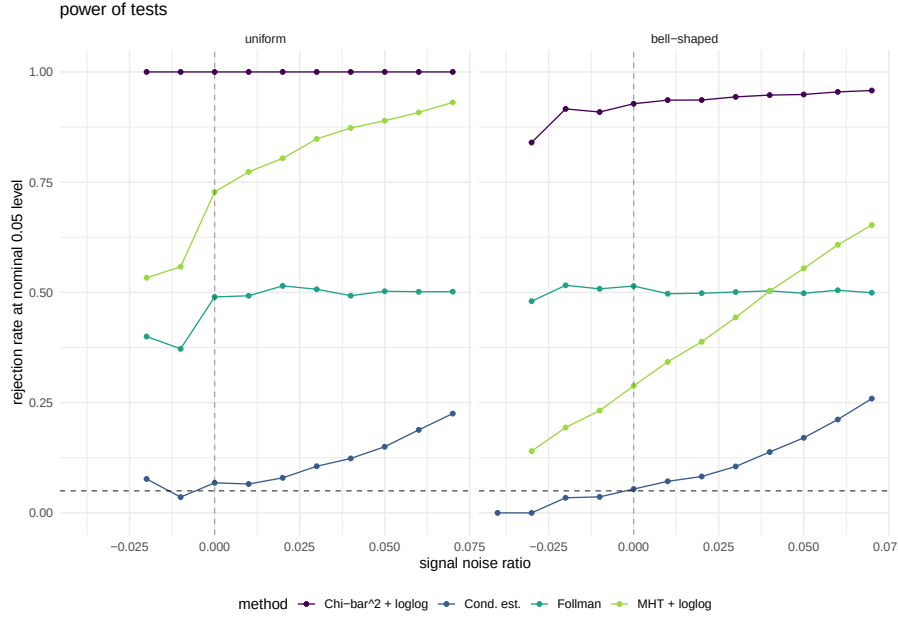


Figure 10: The empirical power of the conditional estimation, Follman’s test, Hansen’s SPA and corrected chi-bar-square procedures are shown versus the signal-noise ratio of the asset with maximum Sharpe ratio for the uniform and bell-shaped arrangements of the vector ζ . We perform 100,000 simulations with $n = 1008$, $k = 105$. Cases with low incidence were removed as the simulated rejection rates are not representative.

We doubt that the simple experiments performed here have revealed all the relevant differences between the various tests or when one dominates the others.

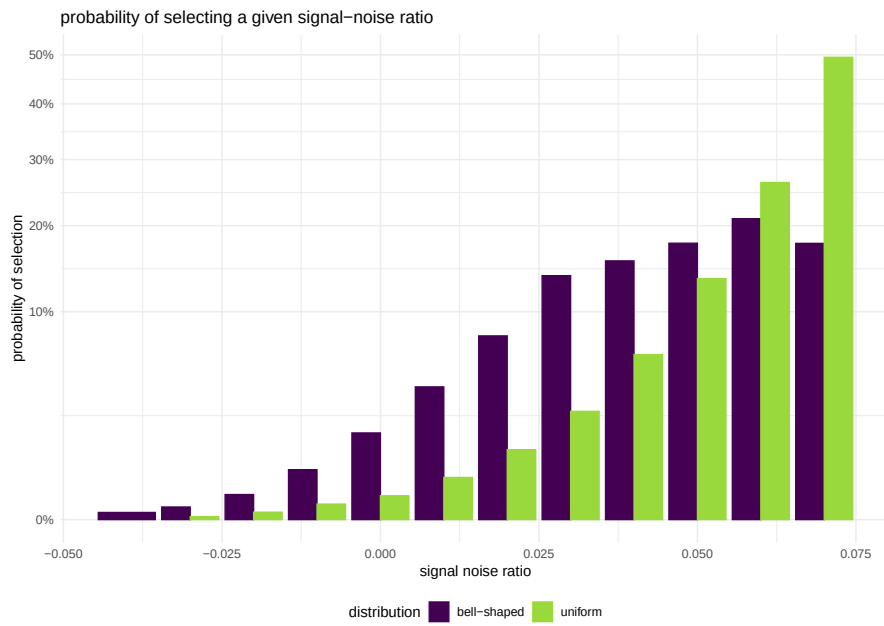


Figure 11: The probability that ζ_1 , the signal-noise ratio of the asset selected for having the maximum Sharpe ratio, takes a given value in our simulations is shown. The y axis is in square root space to show detail.

4.3 Simulations of False Discovery Rate

Recalling the discussion in Section 3 we now perform simulations similar to the above but focus on the positive False Discovery Rate (pFDR). [29] We define the pFDR as the conditional probability of making a false positive conditional on the statistical test rejecting the null. Recalling Figure 2, this is the quantity

$$pFDR = \frac{\gamma_1 + \gamma_3}{\gamma_1 + \gamma_3 + \gamma_5}. \quad (18)$$

Similar to the experiments above we consider a few different configurations for ζ , though by no means do we consider an exhaustive collection. In each configuration we will perform a number of simulations of the different tests considered in this paper. We parametrize the configurations by the maximal signal-noise ratio, ζ_{max} , then we will plot the results against that ζ_{max} . As above we consider:

1. “one-good,” where 1 asset has signal-noise ratio of ζ_{max} , and the remaining $k - 1$ take value of $-\zeta_{max}$.
2. “half-good,” where $k/2$ have signal-noise ratio of ζ_{max} and the remaining take value of $-\zeta_{max}$.
3. “uniform,” where the signal-noise ratios are uniformly distributed across 15 different values from $-\zeta_{max}$ to ζ_{max} , as in Figure 9.
4. “bell-shaped,” where the signal-noise ratios are distributed across a nearly normal distribution spanning from $-\zeta_{max}$ to ζ_{max} , as in Figure 9.

In each configuration, except one-good, the mean signal-noise ratio across the different assets is zero. We perform simulations with $k = 105$, and $n = 1008$. We let ζ_{max} vary from near zero to around $1\text{yr}^{-1/2}$. For each setting we perform 10,000 simulations, testing at the 0.05 level. We compute the empirical pFDR of each technique. When a technique makes no rejections we “nan out” the value and do not plot a marker. This happens for the chi-bar-square test in the one-good configuration for some values of ζ_{max} .

We plot the results in Figure 12. In that plot we also plot the results of using the naïve test for significance of the signal-noise ratio, which we denote as “no MHT”. [24] We also plot the pFDR resulting from always rejecting the null hypothesis, which we plot as “Reject All.” This line shows the background rate of bad strategies, which is $\beta_0 + \beta_1$ in the language of Figure 2. The Reject All line is essentially the worst pFDR that a test could display without doing something perverse.

We see that MHT and the conditional procedure generally have the lowest pFDR, generally lower than the chi-bar-square procedure, lower than the naïve test, and lower than the Reject All benchmark. The MHT performs best in the one-good configuration. We perform some additional simulations under the uniform and bell-shaped configurations, since these seem more realistic than the one-good and half-good. We perform 100,000 simulations for these configurations, plotting the results in Figure 13. There we see that the conditional inference procedure has slightly lower pFDR than the MHT procedure when ζ_{max} is sufficiently large. It is hard to imagine that this result scales to all sample sizes and configurations, but we are generally confident that the two procedures deliver similar pFDR.

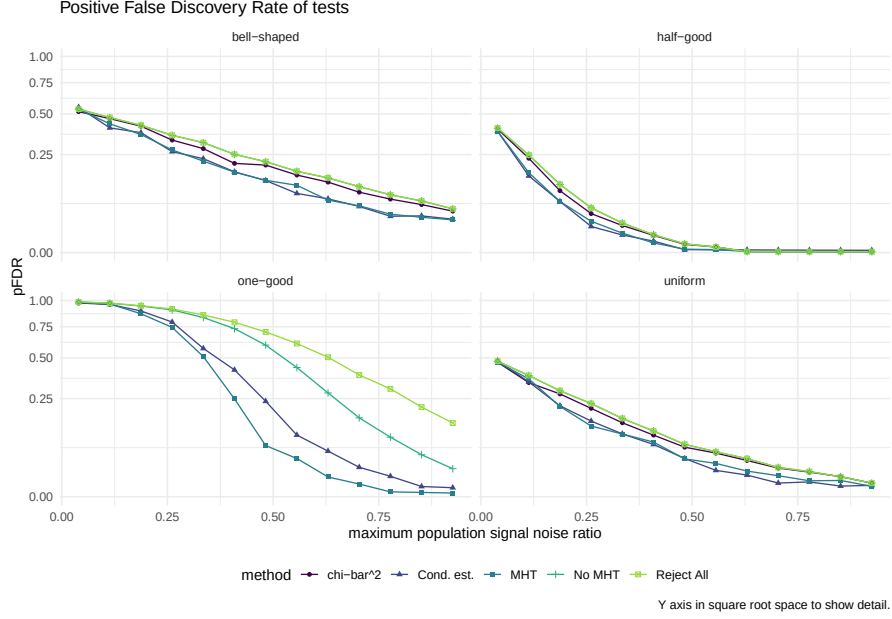


Figure 12: The empirical positive False Discovery Rate (pFDR) of the various tests are shown versus the maximum population signal-noise ratio, ζ_{max} , for various configurations of the true signal-noise ratio. We perform 10,000 simulations with $n = 1008$, $k = 105$. The chi-bar-square test has no rejections for most of the one-good case, and is not plotted there.

4.3.1 Minimizing False Discovery Rate

As a thought experiment, let us consider how we might minimize the pFDR for a toy problem. Let π be the probability that the selected strategy satisfies the null hypothesis, which we define to mean having signal-noise ratio of zero. We have $\pi = \beta_0 + \beta_1$ in the language of Figure 2. Suppose that the type I rate under the null is α . Under the alternative, which occurs with probability $1 - \pi$, the strategy is “good” and has a signal-noise ratio of ζ_{max} . Suppose the observed Sharpe ratio is the signal-noise ratio plus a normal noise with unit variance. So our test consists of rejecting the null precisely when the Sharpe ratio is greater than $\Phi^{-1}(1 - \alpha)$, where $\Phi(x)$ is the cumulative density of the standard normal distribution. Under the alternative this occurs with probability $1 - \Phi(\Phi^{-1}(1 - \alpha) - \zeta_{max})$.

Then the pFDR is

$$\begin{aligned}
 pFDR &= \frac{\pi\alpha}{\pi\alpha + (1 - \pi)(1 - \Phi(\Phi^{-1}(1 - \alpha) - \zeta_{max}))}, \\
 &= \frac{1}{1 + \frac{(1 - \pi)(\Phi(\zeta_{max} - \Phi^{-1}(1 - \alpha)))}{\pi\alpha}}, \\
 &= \frac{1}{1 + \frac{(1 - \pi)(\Phi(\zeta_{max} + \Phi^{-1}(\alpha)))}{\pi\Phi(\Phi^{-1}(\alpha))}}.
 \end{aligned}$$

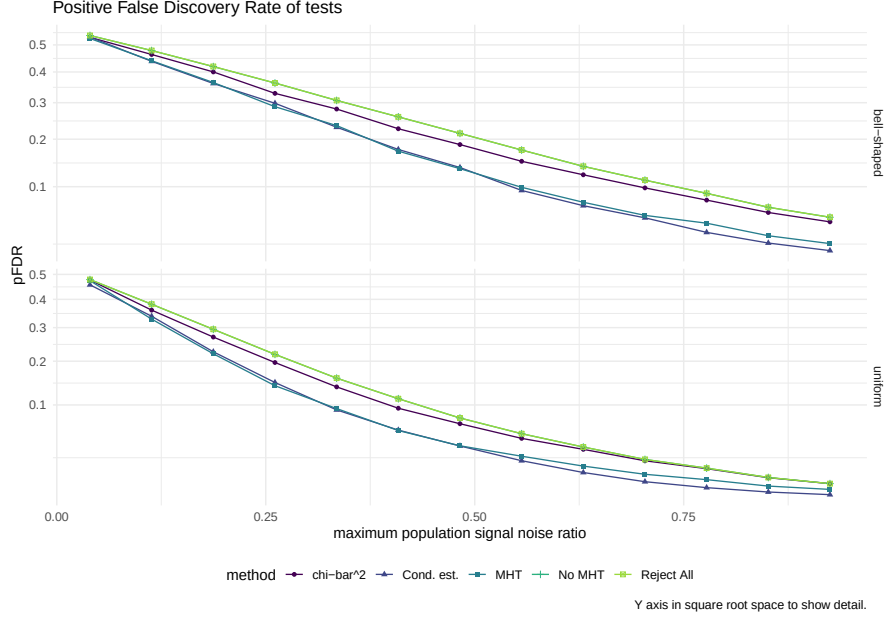


Figure 13: The empirical positive False Discovery Rate (pFDR) of the various tests are shown versus the maximum population signal-noise ratio for various arrangements of the true signal-noise ratio. We perform 100,000 simulations with $n = 1008$, $k = 105$.

Without asking the analyst to improve π by testing only good strategies, to minimize the pFDR we should maximize the quantity

$$\frac{\Phi(\zeta_{max} + \Phi^{-1}(\alpha))}{\Phi(\Phi^{-1}(\alpha))}.$$

Linearizing this with Taylor's theorem we want to approximately maximize the quantity

$$\frac{\Phi(\zeta_{max} + \Phi^{-1}(\alpha))}{\Phi(\Phi^{-1}(\alpha))} \approx 1 + \zeta_{max} \frac{\phi(\Phi^{-1}(\alpha))}{\Phi(\Phi^{-1}(\alpha))}.$$

Note that the quantity $\phi(x)/\Phi(-x)$ is called the *inverse Mills ratio*. It is well known that the inverse Mills ratio is unbounded and increasing as $x \rightarrow \infty$. The implication for our problem is that pFDR is minimized as we drive the type I rate to zero.

While we cannot literally set $\alpha = 0$, since we would never reject the null hypothesis, it is easy to reconsider the experiments above using different type I rates. That is exactly what we do: we take the MHT and conditional test results from the simulations above, wherein we computed their p-values, and compare them against different target type I rates, estimating the pFDR in each case. We plot these in Figure 14. To show detail in this plot, we have divided the pFDR by the “worst case” pFDR one gets from the Reject All test, which is to say the background rate of bad strategies.

We indeed see in this plot that setting the type I rate to 0.005 yields the lowest pFDR, and that the pFDR values are generally monotonic in the targeted

type I rate. We see that at the most extreme values of α that the conditional procedure has slightly lower pFDR than the MHT test, but again this is not an exhaustive study of the possible configurations. We would recommend, however, that if one seeks to minimize pFDR the choice of α is likely more important than which test you apply.

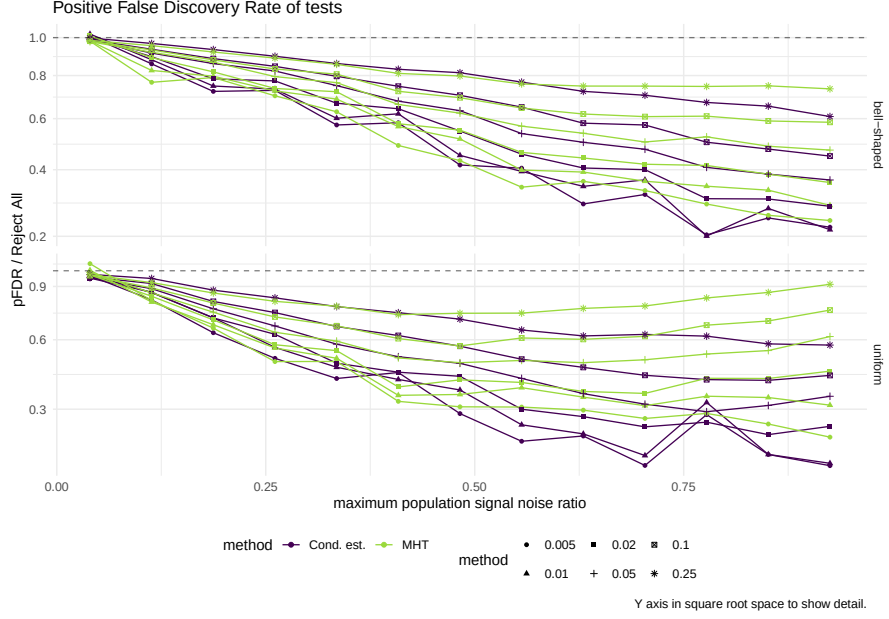


Figure 14: The empirical positive False Discovery Rate (pFDR) of the conditional estimation and MHT procedures, divided by the Reject All pFDR are plotted versus the maximum population signal-noise ratio, ζ_{max} , for various configurations of the true signal-noise ratio. We perform 100,000 simulations with $n = 1008$, $k = 105$.

4.4 Simulations under the null with correlated returns

The naïve MHT test cannot maintain the nominal type I rate in the face of correlated assets. To demonstrate this, we repeat the simulations above, testing the various statistical tests, but with $R = \rho(\mathbf{1}\mathbf{1}^\top) + (1 - \rho)\mathbf{I}$ and $\zeta = \mathbf{0}$. We set $k = 20$, $n = 504$, and perform 10^4 simulations to estimate the empirical rejection rate. In Figure 15, we plot the empirical rejection rate versus ρ at the nominal 0.05 type I level. While the conditional estimation procedure and one-sided tests appear to maintain the nominal rejection rate, the MHT test is conservative, with near zero rejection rates for large ρ . The fix for common correlation described in Section 3.1 is also tested, yielding nominal rejection rates.

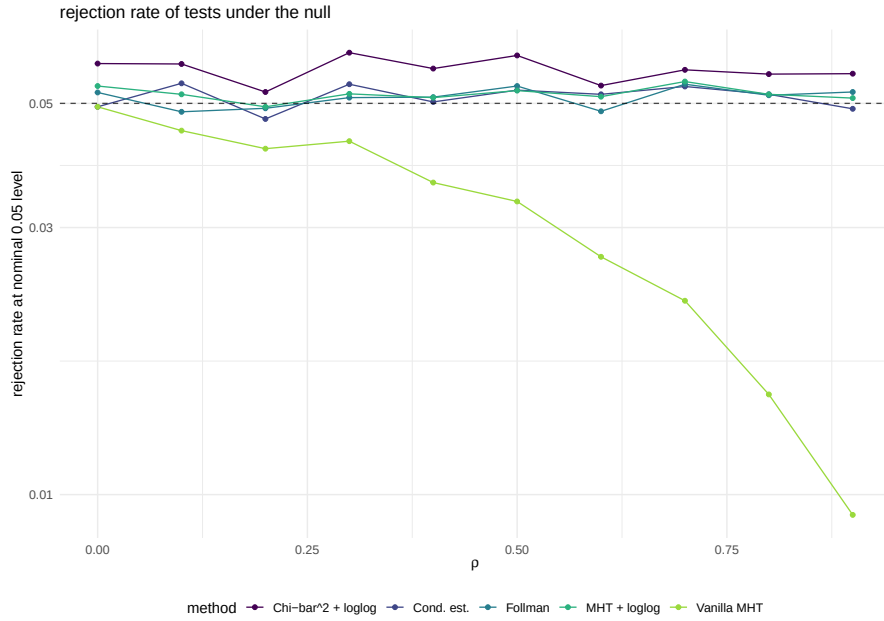


Figure 15: The empirical type I rate under the null hypothesis is plotted versus ρ for the case where $R = \rho(\mathbf{1}\mathbf{1}^\top) + (1 - \rho)\mathbf{I}$, for several testing procedures: the conditional estimation, the chi-bar square test, Follman’s test, the vanilla Bonferroni correction, Hansen’s corrected chi-bar-square and SPA tests, and Bonferroni correction with fix for common correlation. Tests are performed with Gaussian returns, for 20 assets over 504 days. Tests were performed at the 0.05 level, which appears to be maintained by all procedures except the regular Bonferroni procedure, and Hansen’s chi-bar-square test, which has slightly higher than nominal rejection rates for this n . Empirical rates are over 10,000 simulations. The y axis is in log scale.

4.5 Example: Five Industry Portfolios

We download the monthly returns of five industry portfolios from Ken French’s data library. [7] We consider the 1187 months of data on five industries from Jan 1927 through Nov 2025. We compute the Sharpe ratio of the returns of each, and present them in Table 1. We have reordered the industries in decreasing Sharpe ratio. The industry portfolio with the highest Sharpe ratio was Healthcare with a Sharpe ratio of around $0.193 \text{ mo.}^{-1/2}$ which is approximately $0.669 \text{ yr}^{-1/2}$.

industry	Sharpe Ratio
Healthcare	$0.193 \text{ mo.}^{-1/2}$
Consumer	$0.192 \text{ mo.}^{-1/2}$
Technology	$0.181 \text{ mo.}^{-1/2}$
Manufacturing	$0.174 \text{ mo.}^{-1/2}$
Other	$0.144 \text{ mo.}^{-1/2}$

Table 1: The Sharpe ratios of the five industry portfolios are shown.

We are interested in computing 95% upper confidence intervals on the signal-noise ratio of the Healthcare portfolio. We are only considering this portfolio as it is the one with maximum Sharpe ratio in our sample. If we had been interested in testing Healthcare without our conditional selection, we would compute the confidence interval $[0.145 \text{ mo.}^{-1/2}, \infty)$ based on inverting the non-central t -distribution. [22, 24, 21] If instead we approximate the standard error by plugging in $0.193 \text{ mo.}^{-1/2}$ as the signal-noise ratio of Healthcare into Equation 6, we estimate the standard error of the Sharpe ratio to be $0.029 \text{ mo.}^{-1/2}$. Based on this we can compute the naïve confidence interval of the measured Sharpe ratio plus $z_{0.05} = -1.645$ times the standard error. This also gives the confidence interval $[0.145 \text{ mo.}^{-1/2}, \infty)$.

Using the simple Bonferroni correction, however, since we selected Healthcare only for having the maximum Sharpe ratio, we should compute the confidence interval by adding $z_{0.01} = -2.326$ times the standard error. This yields the confidence interval $[0.125 \text{ mo.}^{-1/2}, \infty)$.

The correlation of industry returns is high, however. The pairwise sample correlations range from 0.699 to 0.892 with a median value of 0.793. Plugging this value in as ρ , we find the value c such that the z_1 from Equation 13 is equal to $z_{0.01} = -2.326$. This leads to the confidence interval $[0.13 \text{ mo.}^{-1/2}, \infty)$.

We use this estimate of ρ to compute the chi-bar square test. We invert the test to find the 95% upper confidence interval $[0.146 \text{ mo.}^{-1/2}, \infty)$.

Finally we use the conditional estimation procedure, inverting the hypothesis test to find the corresponding population value. This yields the confidence interval $[-0.056 \text{ mo.}^{-1/2}, \infty)$. This is a much wider interval which encompasses zero. The extra width is not surprising given the results of our simulations, and is the price we pay for testing the selected asset conditionally on the observed.

5 Conclusions and Future Work

The conditional estimation procedure appears to achieve nominal type I rates under the null, and does not seem unduly harmed by assuming the vector $\hat{\zeta}$ is normally distributed. Nor does it seem to suffer greatly from using sample

estimates of the correlation matrix, R , nor from the presence of kurtotic returns. The procedure can be used for other test configurations beyond the asset with the maximum Sharpe ratio, and can be used to construct confidence intervals. It appears to have low power compared to tests based on the MHT for some test configurations. However MHT tests can have very high conditional type I rates in those unlikely cases where an asset with low signal-noise ratio is selected, while the conditional estimation procedure maintains the type I rate in this case.

The low power of the test gives us reason to seek improvements. Perhaps the conditioning procedure can be adapted to recognize that the strategist would have been testing another asset if the Sharpe ratio of the currently selected asset had been lower. However, it would seem that in so doing, one would lose the desirable property of testing the signal-noise ratio of the selected asset, instead of testing the more general hypothesis of (8). Perhaps its power can be increased using Hansen’s log log adjustment.

On the other hand, the conditional procedure appears to have low positive False Discovery Rates, and is competitive with the other tests on this metric. We suggest setting low target type I rates to minimize the pFDR. Perhaps the procedure of Storey can be adapted to this problem to estimate the pFDR for a given type I rate. [29]

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A Establishing Equation 7.

Let $\mathbf{V}$ be the variance covariance to be computed. Then from Equation 2,

$$\begin{aligned}
 \mathbf{V} &= \left(\frac{d\boldsymbol{\zeta}}{d[\boldsymbol{\mu}^\top, \boldsymbol{\alpha}_2^\top]^\top} \right) \Omega \left(\frac{d\boldsymbol{\zeta}}{d[\boldsymbol{\mu}^\top, \boldsymbol{\alpha}_2^\top]^\top} \right)^\top, \\
 &= \text{Diag} \left(\frac{1}{\boldsymbol{\sigma}^2} \right) \begin{bmatrix} \text{Diag}(\boldsymbol{\sigma} + \boldsymbol{\mu}\boldsymbol{\zeta}) & \text{Diag}\left(\frac{-\boldsymbol{\zeta}}{2}\right) \end{bmatrix} \Omega \begin{bmatrix} \text{Diag}(\boldsymbol{\sigma} + \boldsymbol{\mu}\boldsymbol{\zeta}) \\ \text{Diag}\left(\frac{-\boldsymbol{\zeta}}{2}\right) \end{bmatrix} \text{Diag} \left(\frac{1}{\boldsymbol{\sigma}^2} \right).
 \end{aligned}$$

Plugging in Ω from Equation 4, we have

$$\begin{aligned}
V &= \text{Diag} \left(\frac{1}{\sigma^2} \right) \left\{ \text{Diag} (\sigma + \mu\zeta) \Sigma \text{Diag} (\sigma + \mu\zeta) + \text{Diag} (\sigma + \mu\zeta) 2\Sigma \text{Diag} (\mu) \text{Diag} \left(\frac{-\zeta}{2} \right) \right. \\
&\quad + \text{Diag} \left(\frac{-\zeta}{2} \right) 2 \text{Diag} (\mu) \Sigma \text{Diag} (\sigma + \mu\zeta) + \text{Diag} \left(\frac{-\zeta}{2} \right) 2\Sigma \odot \Sigma \text{Diag} \left(\frac{-\zeta}{2} \right) \\
&\quad \left. + \text{Diag} \left(\frac{-\zeta}{2} \right) 4 \text{Diag} (\mu) \Sigma \text{Diag} (\mu) \text{Diag} \left(\frac{-\zeta}{2} \right) \right\} \text{Diag} \left(\frac{1}{\sigma^2} \right), \\
&= \text{Diag} \left(\frac{1}{\sigma^2} \right) \left\{ \text{Diag} (\sigma) \Sigma \text{Diag} (\sigma) - \text{Diag} (\mu\zeta) \Sigma \text{Diag} (\mu\zeta) \right. \\
&\quad + \text{Diag} \left(\frac{-\zeta}{2} \right) 2\Sigma \odot \Sigma \text{Diag} \left(\frac{-\zeta}{2} \right) \\
&\quad \left. + \text{Diag} (\mu\zeta) \Sigma \text{Diag} (\mu\zeta) \right\} \text{Diag} \left(\frac{1}{\sigma^2} \right), \\
&= \text{Diag} \left(\frac{1}{\sigma^2} \right) \left\{ \text{Diag} (\sigma) \Sigma \text{Diag} (\sigma) + \frac{1}{2} \text{Diag} (\zeta) \Sigma \odot \Sigma \text{Diag} (\zeta) \right\} \text{Diag} \left(\frac{1}{\sigma^2} \right), \\
&= R + \frac{1}{2} \text{Diag} (\zeta) \text{Diag} \left(\frac{1}{\sigma^2} \right) \Sigma \odot \Sigma \text{Diag} \left(\frac{1}{\sigma^2} \right) \text{Diag} (\zeta), \\
&= R + \frac{1}{2} \text{Diag} (\zeta) R \odot R \text{Diag} \left(\frac{1}{\sigma^2} \right).
\end{aligned}$$

Proving Equation 7 is similar, and is left as an exercise for the reader.